

Greenfield Municipal Center

CMVP Capstone Project — Offline Data Reference Package

Building Scenario

Parameter	Value
Facility	Greenfield Municipal Center — mixed-use government building
Location / Climate	Mid-Atlantic region (heating-dominant, significant summer cooling)
Total Floor Area	62,000 sq ft
Contract Type	Energy Savings Performance Contract (ESPC)
Baseline Period	Year 1 — 12 months (January through December)
Reporting Period	Year 2 — 12 months (January through December)
Electricity Rate	\$0.105 / kWh (flat blended rate)
Natural Gas Rate	\$1.15 / therm
Elec CO ₂ Factor	0.417 kg CO ₂ /kWh (mid-Atlantic grid average)
Gas CO ₂ Factor	5.302 kg CO ₂ /therm

ECM Summary

ECM	Description	M&V Approach	Type
ECM-1	LED lighting + occupancy controls (Wings A & B)	Retrofit Isolation	Stipulated (cf. Option A)
ECM-2	Chiller replacement (central plant)	Whole Facility or Retrofit Isolation	Student Decision
ECM-3	Building envelope — roof insulation	Whole Facility	cf. Option C
ECM-4	VFDs on AHU supply fans (3 units)	Retrofit Isolation	Continuous (cf. Option B)

Wings & Spaces

Wing / Area	Use	Area (sq ft)	Schedule
Wing A	Office space	35,000	Mon–Fri, standard hours
Wing B	Public library & community rooms	15,000	Extended hours incl. evenings
Wing C	Data center / IT operations	2,000	24/7 constant load
Common Areas	Lobby, corridors, mechanical	10,000	Varies

This package contains the complete Greenfield dataset for use without internet access. All data matches the web application exactly. Tabs: Baseline Data | Reporting Data | Model Results | Lighting Calc | Fan Law Calc | Savings Calc.

Baseline Monthly Data

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Baseline Period — Monthly Utility Data

Pre-retrofit · 12 months · No ECMs installed · Total kWh and Total Therms are the whole-facility dependent variables for model fitting.

Month	Days	Avg OAT (°F)	Total kWh	Total Therms	Lighting (kWh)	Cooling (kWh)	Fan (kWh)	Plug (kWh)	Data Ctr (kWh)	Htg (therms)	DHW (therms)
Jan	31	33.0	108,662	1,139	14,904	10,061	11,002	14,140	37,274	889	250
Feb	28	34.9	93,429	779	13,225	9,114	9,442	12,598	33,667	545	234
Mar	31	45.9	93,584	482	14,293	11,136	9,997	13,692	37,274	338	145
Apr	30	54.4	89,039	343	14,345	12,492	9,829	13,626	36,072	240	103
May	31	64.7	95,784	270	14,904	18,641	10,233	14,140	37,274	54	216
Jun	30	72.5	95,115	223	13,734	22,395	9,676	13,178	36,072	45	179
Jul	31	76.1	104,702	237	14,904	28,107	10,270	14,140	37,274	47	190
Aug	31	73.4	100,216	237	14,682	24,007	10,242	13,970	37,274	47	189
Sep	30	68.2	92,868	236	13,956	19,542	9,693	13,347	36,072	47	189
Oct	31	53.5	92,756	366	14,904	13,037	10,174	14,140	37,274	256	110
Nov	30	46.1	90,863	488	14,123	10,245	9,838	13,456	36,072	342	146
Dec	31	33.8	105,610	1,001	14,515	9,915	10,581	13,861	37,274	751	250
TOTAL	365	54.7	1,162,628	5,801	172,489	188,692	120,977	164,288	438,873	3,601	2,201

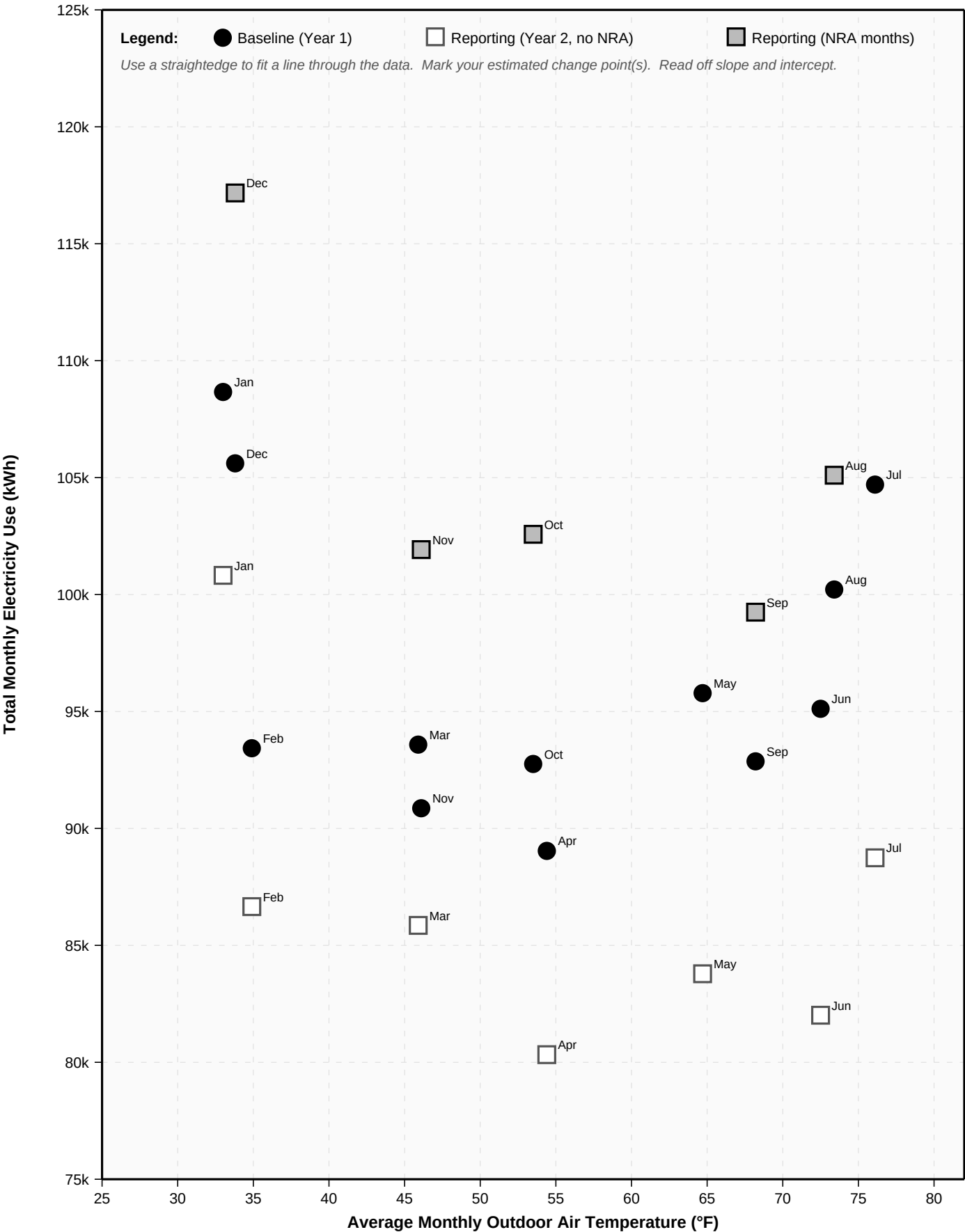
NOTE: Lighting, Cooling, Fan, Plug, and Data Center sub-columns are provided for educational reference only. In a real project these would not be available from utility bills — you would only have Total kWh and Total Therms from the utility meter.

Scatter Plot Analysis Questions — Complete Before Model Fitting

1. Plot Avg OAT (x) vs Total kWh (y) for the 12 baseline months. Describe the shape.
2. At approximately what OAT does electricity use appear to reach a minimum?
3. Plot Avg OAT (x) vs Total Therms (y). Describe the relationship.
4. At approximately what OAT does gas use drop to zero? What does this represent physically?
5. Which end-use explains the heating-season electricity increase? Which explains the cooling peak?

Scatter Plot A — Average OAT vs. Total Electricity (kWh)

Baseline and Reporting periods · Use a straightedge to fit a change-point model · Identify approximate heating CP and cooling CP

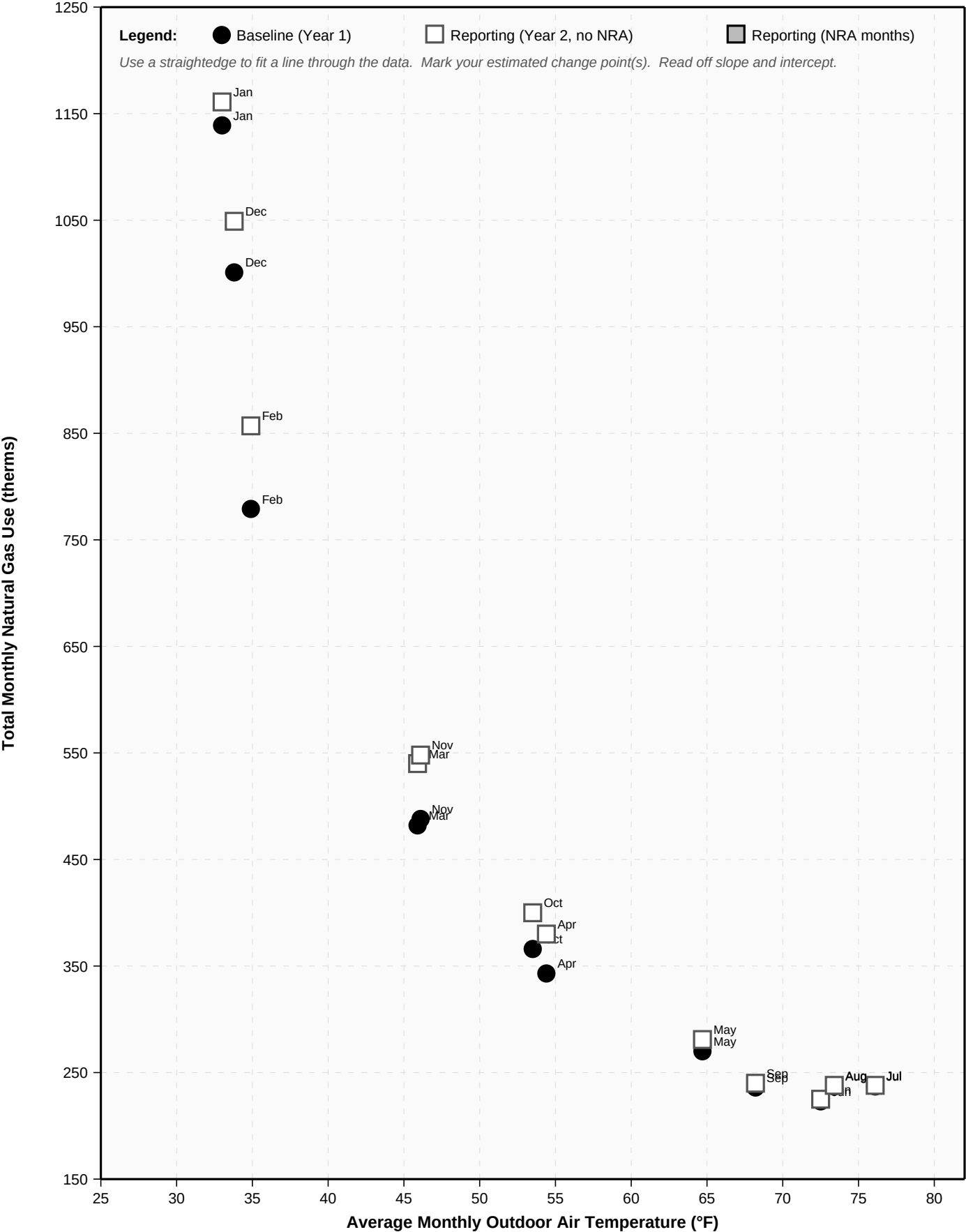


Reference values at minimum (balance point ~55-65°F): Baseline ≈ 90,000-93,000 kWh. Heating season low point: ~33°F (Jan/Dec) → ~105,000-109,000 kWh. Cooling season high point: ~76°F (Jul) → ~88,000-90,000 kWh.

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
OAT(°F)	33	35	46	54	65	72	76	73	68	54	46	34
kWh(B)	108662	93429	93584	89039	95784	95115	104702	100216	92868	92756	90863	105610
kWh(R)	100821	86657	85854	80327	83786	82013	88744	105103	99247	102576	101922	117172

Scatter Plot B — Average OAT vs. Total Natural Gas (therms)

Baseline and Reporting periods · Use a straightedge to fit a 3-parameter heating model · Identify the heating change point (OAT where gas use goes to baseload)



At high OAT (summer months), gas approaches a constant baseload (DHW only). The heating change point is the OAT where the line departs from that baseload. Estimate

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
OAT(°F)	33	35	46	54	65	72	76	73	68	54	46	34
Thm(B)	1139	779	482	343	270	223	237	237	236	366	488	1001
Thm(R)	1161	857	540	380	281	225	238	238	240	400	548	1049

Reporting Monthly Data

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Reporting Period — Monthly Utility Data

Post-retrofit · 12 months · All ECMS installed · NRA Event: Data center expansion begins Month 8 (August) · Shaded rows = NRA-affected months.

Month	Days	Avg OAT (°F)	Total kWh	Total Therms	Lighting (kWh)	Cooling (kWh)	Fan (kWh)	Plug (kWh)	Data Ctr (kWh)	Htg (therms)	DHW (therms)
Jan	31	33.0	100,821	1,161	9,664	9,957	7,257	14,140	37,274	911	250
Feb	28	34.9	86,657	857	8,593	9,007	6,406	12,598	33,667	607	250
Mar	31	45.9	85,854	540	9,315	10,566	6,898	13,692	37,274	378	162
Apr	30	54.4	80,327	380	9,307	11,171	6,758	13,626	36,072	266	114
May	31	64.7	83,786	281	9,664	14,872	7,022	14,140	37,274	56	225
Jun	30	72.5	82,013	225	8,958	17,024	6,680	13,178	36,072	45	180
Jul	31	76.1	88,744	238	9,664	20,615	7,040	14,140	37,274	48	190
Aug ■ NRA	31	73.4	105,103	238	10,433	21,526	7,026	13,903	52,154	48	191
Sep ■ NRA	30	68.2	99,247	240	9,938	18,509	6,690	13,282	50,472	48	192
Oct ■ NRA	31	53.5	102,576	400	10,553	14,910	6,990	14,072	52,154	280	120
Nov ■ NRA	30	46.1	101,922	548	10,048	13,081	6,754	13,391	50,472	384	164
Dec ■ NRA	31	33.8	117,172	1,049	10,323	13,129	7,092	13,794	52,154	799	250
TOTAL 365G		54.7	1,134,222	6,157	116,460	174,367	82,613	163,956	512,313	3,870	2,288

NRA EVENT — Data Center Expansion (Teaching Moment 3)

Detection: Step change in electric consumption visible starting Month 8. Total kWh jumps ~19,000 kWh above the baseline model prediction despite t

Cause: Wing C data center expanded — additional server racks added, increasing constant electrical load from ~37,274 kWh/month to ~52,154 kWh

Without adjustment: Savings appear negative in Months 8–12. With NRA applied: Retrofit savings are properly isolated and reported.

Question: What evidence would you require to support this NRA? Who must agree to the adjustment before it can be applied?

Model Results

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Baseline Model Results — Pre-Computed Statistics

5-Parameter Change-Point Model (electric) · 3-Parameter Heating Model (gas) · ASHRAE Guideline 14 validation thresholds: $CV(RMSE) \leq 15\%$, $NMBE \leq 5\%$

Electric Model (5-Parameter Change-Point)

Parameter	Physical / Constrained Model	Auto-Fit / Statistical Model
Model type	5P Change-Point (H+C)	5P Change-Point (H+C)
Heating change point (CP1)	48 °F	35 °F
Cooling change point (CP2)	63 °F	71 °F
Baseline intercept	37,274 kWh/month	37,274 kWh/month
Heating slope	-2,087 kWh/°F·month	-2,087 kWh/°F·month
Cooling slope	+1,664 kWh/°F·month	+1,664 kWh/°F·month
R ²	0.91	0.95
CV(RMSE)	5.1%	3.8%
NMBE	0.3%	0.3%
G14 CV(RMSE) $\leq 15\%$?	✓ PASS	✓ PASS
G14 NMBE $\leq 5\%$?	✓ PASS	✓ PASS

★ Teaching Moment 1 — Physics vs. Statistics

The auto-fit algorithm minimizes residuals mathematically, finding change points at 35/71°F ($R^2 = 0.95$). But 35°F as a cooling balance point is physically unrealistic.

The physically constrained model (48/63°F, $R^2 = 0.91$) sacrifices 4 points of R^2 for an interpretable result. Which would you report in an M&V plan? How would you justify the choice?

Gas Model (3-Parameter Heating)

Parameter	Value	G14 Status
Model type	3P Change-Point (heating only)	
Heating change point	57 °F	
Baseline intercept (DHW baseload)	47 therms/month	
Heating slope	-36.1 therms/°F·month	
R ²	0.97	
CV(RMSE)	16.1% ← EXCEEDS 15% THRESHOLD	✗ FAIL
NMBE	1.2%	✓ PASS

★ Teaching Moment 2 — Guideline 14 Failure: Is the Model Unusable?

$CV(RMSE) = 16.1\%$ exceeds the G14 threshold of 15%. $R^2 = 0.97$ is excellent. $NMBE = 1.2\%$ passes easily. The model captures the shape perfectly.

Discussion: Does failing CV(RMSE) by 1 point mean the model is useless? What options does the M&V professional have? What would you document?

ECM-1 Lighting Stipulation

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ECM-1 — Lighting Stipulation Calculator

Key Parameter Measurement (cf. IPMVP Option A) · Fixture wattage verified from spec sheets · Annual operating hours stipulated from building schedules

Space	Qty	Baseline (W)	Retrofit (W)	ΔW	Annual Hours	kWh Saved (calculated)
Wing A — Open Office (2nd fl.)	120	128	40	88	2860	30,201
Wing A — Open Office (1st fl.)	120	128	40	88	2860	30,201
Wing A — Private Offices	80	96	32	64	2600	13,312
Wing A — Conference Rooms	40	128	40	88	1200	4,224
Wing A — Corridors & Restrooms	60	64	20	44	3380	8,923
Wing B — Reading Rooms	100	128	40	88	3900	34,320
Wing B — Stacks	150	96	32	64	2600	24,960
Wing B — Meeting Rooms	30	128	40	88	1500	3,960
Wing B — Circulation & Entry	40	64	20	44	4160	7,321
TOTAL	740					157,424

Annual Hours are stipulated values based on building operating schedules. Typical ranges: offices 2,500–3,000 hrs/yr | library/extended 3,500–4,500 hrs/yr | corridors (near-continuous) 3,000–4,000 hrs/yr. What is your documented basis for each value?

★ Teaching Moment 4 — Interactive Effects

This stipulation calculates ECM-1 savings in isolation: 157,424 kWh/yr. The whole-facility model will show a different total because reducing lighting w

Question: Would you expect the whole-facility model to show more or fewer kWh saved than the lighting stipulation alone? In which season would the

ECM-4 — VFD Fan Law Reference & Spot Measurements

Continuous Performance Verification (cf. IPMVP Option B) · Affinity Law: $P_{\text{new}}/P_{\text{old}} = (N_{\text{new}}/N_{\text{old}})^3$ where P = power, N = speed

Fan Law Quick Reference

Speed Reduction	New Speed (%)	Fan Law Factor (N/100) ³	Power Reduction (%)
10%	90%	0.729	27.1%
20%	80%	0.512	48.8%
30%	70%	0.343	65.7%
35%	65%	0.274	72.6%
40%	60%	0.216	78.4%
50%	50%	0.125	87.5%

Shaded row (35% speed reduction) = design operating point for Greenfield AHUs.

AHU Nameplate Data (Pre-VFD Baseline)

Unit	Design Power (kW)	Design CFM	Location / Zone
AHU-1	8.85 kW	15,000 CFM	Wing A supply
AHU-2	4.72 kW	8,000 CFM	Wing B / Wing C supply
AHU-3	2.68 kW	5,000 CFM	Common areas supply
TOTAL	16.25 kW	28,000 CFM	

Sample Spot Measurements — March 15 Operating Hours (Pre vs. Post VFD)

Hour	OAT (°F)	AHU-1 Pre (kW)	AHU-1 Post (kW)	AHU-1 Speed	AHU-2 Pre (kW)	AHU-2 Post (kW)	AHU-3 Pre (kW)	AHU-3 Post (kW)
08:00	29.9	8.85	4.42	100%	4.72	2.36	2.68	1.33
09:00	37.7	8.85	3.93	89%	4.72	2.10	2.68	1.21
10:00	43.1	8.85	3.36	76%	4.72	1.79	2.68	1.06
11:00	47.4	8.85	2.89	65%	4.72	1.54	2.68	0.95
12:00	50.3	8.85	2.59	59%	4.72	1.38	2.68	0.87
13:00	51.1	8.85	2.50	57%	4.72	1.34	2.68	0.85
14:00	53.6	8.85	2.24	51%	4.72	1.20	2.68	0.78
15:00	55.6	8.85	2.02	46%	4.72	1.08	2.68	0.73

★ Teaching Moment 5 — Fan Law in Practice

AHU-1 at 09:00: Fan law predicts $0.89^3 = 0.705 \rightarrow 8.85 \times 0.705 = 6.23$ kW. Actual measured post-VFD power = 3.93 kW — significantly lower. Why?

Key insight: Small errors in speed measurement propagate cubically into power errors. A 5% speed error becomes a ~15% power error. This is why c

Savings Calculator

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Savings Calculator — Whole-Facility Method

Complete model fitting (Model Results page) before using this table · Predicted kWh = 5P model · Predicted Therms = 3P model · Electricity: \$0.105/kWh | Gas: \$1.15/therm

Month	Actual kWh	Predicted kWh (model)	NRA Adj (kWh)	Elec Savings (kWh)	Actual Therms	Predicted Therms	Gas Savings (therms)
Jan	100,821	68,579	—	-32,242	1,161	913	-248
Feb	86,657	64,613	—	-22,043	857	844	-12
Mar	85,854	41,656	—	-44,197	540	447	-92
Apr	80,327	37,274	—	-43,053	380	140	-239
May	83,786	40,102	—	-43,683	281	47	-234
Jun	82,013	53,082	—	-28,931	225	47	-178
Jul	88,744	59,072	—	-29,671	238	47	-191
Aug ■	105,103	54,579	14,880	-35,643	238	47	-191
Sep ■	99,247	45,926	14,880	-38,440	240	47	-193
Oct ■	102,576	37,274	14,880	-50,422	400	173	-227
Nov ■	101,922	41,239	14,880	-45,802	548	440	-108
Dec ■	117,172	66,909	14,880	-35,382	1,049	884	-164
ANNUAL	1,134,222	610,309	74,400	-449,512	6,157	4,080	-2077

Shaded rows = NRA-affected months (data center expansion + 14,880 kWh/month).

Savings Valuation & CO₂

Item	Quantity	Rate	Annual Value
Electricity savings	-449,512 kWh	\$0.105/kWh	\$-47,199
Gas savings	-2077 therms	\$1.15/therm	\$-2,388
TOTAL COST SAVINGS			\$-49,587
Electricity CO ₂ avoidance	-449,512 kWh	0.417 kg/kWh	-187.4 metric tons
Gas CO ₂ avoidance	-2077 therms	5.302 kg/therm	-11.0 metric tons
TOTAL CO₂ AVOIDANCE			-198.5 metric tons

★ Teaching Moment 6 — Gas Savings Despite Higher Consumption

Notice the gas savings column: the counterfactual (predicted therms) is HIGHER than actual therms in months 1, 2, and 12 — meaning the model predicted higher gas use than what actually occurred.

How can ECM-3 (roof insulation) show gas savings when overall gas use went up? The counterfactual isolates ECM performance from weather variations.